hw1

April 5, 2026

1 HW1: Logistic Regression

This class is about models and algorithms for discrete data. This homework will have all 3 ingredients: * **Data**: the results from all college football games in the 2023 season * **Model**: The *Bradely-Terry* model for predicting the winners of football game. The Bradley-Terry model is just logistic regression. * **Algorithm**: We will implement two ways of fitting logistic regression: gradient descent and Newton's method

1.1 The Bradley-Terry Model

In the Bradley-Terry Model, we give team k a team-effect β_k. Basically, higher β_k (relatively speaking), means that team k is a better team. The Bradley-Terry model formalizes this intution by modeling the log odds of team k beating team k' by the difference in their team effects, $\beta_k - \beta_{k'}$.

Let $i = 1, \dots, n$ index games, and let $h(i) \in \{1, \dots, K\}$ and $a(i) \in \{1, \dots, K\}$ denote the indices of the home and away teams, respectively. Let $Y_i \in \{0, 1\}$ denote whether the home team won. Under the Bradley-Terry model,

$$Y_i \sim \text{Bern}(\sigma(\beta_{h(i)} - \beta_{a(i)})),$$

where $\sigma(\cdot)$ is the sigmoid function. We can view this model as a logistic regression model with covariates $x_i \in \mathbb{R}^K$ where,

$$x_{i,k} = \begin{cases} +1 & \text{if } h(i) = k \\ -1 & \text{if } a(i) = k \\ 0 & \text{o.w.,} \end{cases}$$

and parameters $\beta \in \mathbb{R}^K$.

1.2 Data

We use the results of college football games in the fall 2023 season, which are available from the course github page and loaded for you below.

The data comes as a list of the outcomes of individual games. You'll need to wrangle the data to get it into a format that you can feed into the Bradley-Terry model.

[1]:
```
import torch
import torch.nn as nn
import torch.optim as optim
import pandas as pd
import matplotlib.pyplot as plt
```

```python
from tqdm import tqdm
import numpy as np
```

[2]:
```python
allgames = pd.read_csv("01_allgames.csv")
```

[3]:
```python
allgames.head().T
```

[3]:
```
                                          0                         1  \
Id                                401550883                 401525434
Season                                 2023                      2023
Week                                      1                         1
Season Type                           regular                   regular
Start Date            2023-08-26T17:00:00.000Z  2023-08-26T18:30:00.000Z
Start Time Tbd                          False                     False
Completed                                True                      True
Neutral Site                            False                      True
Conference Game                         False                     False
Attendance                                NaN                   49000.0
Venue Id                               5910.0                    3504.0
Venue                        Callaway Stadium             Aviva Stadium
Home Id                                  548                        87
Home Team                    LaGrange College                Notre Dame
Home Conference                     USA South          FBS Independents
Home Division                             iii                       fbs
Home Points                               NaN                      42.0
Home Line Scores                          NaN                       NaN
Home Post Win Prob                        NaN                  0.998958
Home Pregame Elo                          NaN                    1733.0
Home Postgame Elo                         NaN                    1819.0
Away Id                                125762                      2426
Away Team          Florida Memorial University                      Navy
Away Conference                           NaN         American Athletic
Away Division                             NaN                       fbs
Away Points                               NaN                       3.0
Away Line Scores                          NaN                       NaN
Away Post Win Prob                        NaN                  0.001042
Away Pregame Elo                          NaN                    1471.0
Away Postgame Elo                         NaN                    1385.0
Excitement Index                          NaN                  1.346908
Highlights                                NaN                       NaN
Notes                                     NaN                       NaN

                                          2                         3  \
Id                                401540199                 401520145
Season                                 2023                      2023
Week                                      1                         1
Season Type                           regular                   regular
```

Start Date	2023-08-26T19:30:00.000Z	2023-08-26T21:30:00.000Z
Start Time Tbd	False	False
Completed	True	True
Neutral Site	True	False
Conference Game	False	True
Attendance	NaN	17982.0
Venue Id	3689.0	3662.0
Venue	Cramton Bowl	JSU Stadium
Home Id	2382	55
Home Team	Mercer	Jacksonville State
Home Conference	Southern	Conference USA
Home Division	fcs	fbs
Home Points	17.0	17.0
Home Line Scores	NaN	NaN
Home Post Win Prob	0.974151	0.408001
Home Pregame Elo	NaN	1500.0
Home Postgame Elo	NaN	1499.0
Away Id	2453	2638
Away Team	North Alabama	UTEP
Away Conference	UAC	Conference USA
Away Division	fcs	fbs
Away Points	7.0	14.0
Away Line Scores	NaN	NaN
Away Post Win Prob	0.025849	0.591999
Away Pregame Elo	NaN	1369.0
Away Postgame Elo	NaN	1370.0
Excitement Index	6.896909	6.821333
Highlights	NaN	NaN
Notes	NaN	NaN

	4
Id	401525450
Season	2023
Week	1
Season Type	regular
Start Date	2023-08-26T23:00:00.000Z
Start Time Tbd	False
Completed	True
Neutral Site	False
Conference Game	False
Attendance	15356.0
Venue Id	3601.0
Venue	Aggie Memorial Stadium
Home Id	166
Home Team	New Mexico State
Home Conference	Conference USA
Home Division	fbs

```
Home Points                            30.0
Home Line Scores                        NaN
Home Post Win Prob                 0.239249
Home Pregame Elo                     1322.0
Home Postgame Elo                    1274.0
Away Id                                 113
Away Team                             UMass
Away Conference            FBS Independents
Away Division                           fbs
Away Points                            41.0
Away Line Scores                        NaN
Away Post Win Prob                 0.760751
Away Pregame Elo                     1074.0
Away Postgame Elo                    1122.0
Excitement Index                   5.311493
Highlights                              NaN
Notes                                   NaN
```

1.3 Problem 0: Preprocessing

Preprocess the data to drop games with nan scores, construct the covariate matrix X, construct the response vector y, and do any other preprocessing you find useful.

[4]:
```
allgames.values
```

[4]:
```
array([[401550883, 2023, 1, …, nan, nan, nan],
       [401525434, 2023, 1, …, 1.3469076611, nan, nan],
       [401540199, 2023, 1, …, 6.8969089314, nan, nan],
       …,
       [401539478, 2023, 14, …, 8.5801041578, nan, 'ACC Championship'],
       [401539480, 2023, 14, …, 2.7196474905, nan,
        'Big Ten Championship'],
       [401520445, 2023, 15, …, 5.136619754, nan, nan]],
      shape=(3599, 33), dtype=object)
```

[5]:
```
# Frist, We drop games with NaN scores.
df = allgames.dropna(subset=["Home Points", "Away Points", "Neutral Site"])
# Drop Bias(Home Points == Score Points)
df = df[df["Home Points"] != df["Away Points"]]
# Drop Bias(Neutral Site == True)
df = df[df["Neutral Site"] != True] # If we not add line, then Our calculation␣
↪results will be unstable.

# Second, We Construct the response vector $y$.
Y = torch.tensor((df["Home Points"] > df["Away Points"]).astype(int).values,␣
↪dtype=torch.float32).unsqueeze(1)
```

```python
# Last, We Construct the covariate matrix $X$.
teams = sorted(set(df["Home Team"]) | set(df["Away Team"])) # Deduplication
team_idx = {team: i for i, team in enumerate(teams)}
T = len(teams) # Team T

X = torch.zeros(len(df), T) # Design Matrix

# We need to organize the data into a format suitable for Bradley-Terry Model.
df = df.reset_index(drop=True)
for k, r in df.iterrows():
  i = team_idx[r["Home Team"]]
  j = team_idx[r["Away Team"]]
  X[k, i] = 1
  X[k, j] = -1
```

[6]:
```python
df.head().T
```

[6]:

	0	1	\
Id	401520145	401525450	
Season	2023	2023	
Week	1	1	
Season Type	regular	regular	
Start Date	2023-08-26T21:30:00.000Z	2023-08-26T23:00:00.000Z	
Start Time Tbd	False	False	
Completed	True	True	
Neutral Site	False	False	
Conference Game	True	False	
Attendance	17982.0	15356.0	
Venue Id	3662.0	3601.0	
Venue	JSU Stadium	Aggie Memorial Stadium	
Home Id	55	166	
Home Team	Jacksonville State	New Mexico State	
Home Conference	Conference USA	Conference USA	
Home Division	fbs	fbs	
Home Points	17.0	30.0	
Home Line Scores	NaN	NaN	
Home Post Win Prob	0.408001	0.239249	
Home Pregame Elo	1500.0	1322.0	
Home Postgame Elo	1499.0	1274.0	
Away Id	2638	113	
Away Team	UTEP	UMass	
Away Conference	Conference USA	FBS Independents	
Away Division	fbs	fbs	
Away Points	14.0	41.0	
Away Line Scores	NaN	NaN	
Away Post Win Prob	0.591999	0.760751	
Away Pregame Elo	1369.0	1074.0	

Away Postgame Elo	1370.0	1122.0
Excitement Index	6.821333	5.311493
Highlights	NaN	NaN
Notes	NaN	NaN

	2	3 \
Id	401532392	401540628
Season	2023	2023
Week	1	1
Season Type	regular	regular
Start Date	2023-08-26T23:00:00.000Z	2023-08-26T23:00:00.000Z
Start Time Tbd	False	False
Completed	True	True
Neutral Site	False	False
Conference Game	False	False
Attendance	23867.0	NaN
Venue Id	7311.0	4419.0
Venue	Snapdragon Stadium	Bob Ford Field
Home Id	21	399
Home Team	San Diego State	Albany
Home Conference	Mountain West	CAA
Home Division	fbs	fcs
Home Points	20.0	34.0
Home Line Scores	NaN	NaN
Home Post Win Prob	0.954469	0.922517
Home Pregame Elo	1488.0	NaN
Home Postgame Elo	1497.0	NaN
Away Id	195	2230
Away Team	Ohio	Fordham
Away Conference	Mid-American	Patriot
Away Division	fbs	fcs
Away Points	13.0	13.0
Away Line Scores	NaN	NaN
Away Post Win Prob	0.045531	0.077483
Away Pregame Elo	1482.0	NaN
Away Postgame Elo	1473.0	NaN
Excitement Index	6.547378	5.608758
Highlights	NaN	NaN
Notes	NaN	NaN

	4
Id	401520147
Season	2023
Week	1
Season Type	regular
Start Date	2023-08-26T23:30:00.000Z
Start Time Tbd	False

```
Completed                                True
Neutral Site                            False
Conference Game                         False
Attendance                            21407.0
Venue Id                               3973.0
Venue                      Vanderbilt Stadium
Home Id                                   238
Home Team                          Vanderbilt
Home Conference                           SEC
Home Division                             fbs
Home Points                              35.0
Home Line Scores                          NaN
Home Post Win Prob                   0.180846
Home Pregame Elo                       1322.0
Home Postgame Elo                      1327.0
Away Id                                    62
Away Team                             Hawai'i
Away Conference                 Mountain West
Away Division                             fbs
Away Points                              28.0
Away Line Scores                          NaN
Away Post Win Prob                   0.819154
Away Pregame Elo                       1246.0
Away Postgame Elo                      1241.0
Excitement Index                     5.282033
Highlights                                NaN
Notes                                     NaN
```

[7]:
```
X.shape, Y.shape
```

[7]:
```
(torch.Size([3469, 691]), torch.Size([3469, 1]))
```

1.4 Problem 1: Loss function

Write a function to compute the loss, $L(\beta)$ defined be

$$L(\beta) = -\frac{1}{n}\sum_{i=1}^{n} \log p(y_i \mid x_i; \beta) + \frac{\gamma}{2}\|\beta\|_2^2$$

where γ is a hyperparameter that controls the strength of your ℓ_2 regularization.

You may want to use the `torch.distributions.Bernoulli` class.

[8]:
```
def compute_the_loss(x, y, beta, gamma):
  avg_log_likelihood = torch.distributions.Bernoulli(probs=torch.sigmoid(x @␣
 ↪beta)).log_prob(y).mean()
  return -1 * avg_log_likelihood + gamma / 2 * (torch.norm(beta, 2)**2)
```

1.5 Problem 2: Gradient Descent

1.5.1 Problem 2.1 Implementing and checking your gradients

Write a function to compute the gradient of the average negative log likelihood and check your output against the results obtained by PyTorch's automatic differentiation functionality.

Answer:

$$\begin{aligned}\nabla L(\beta) &= \nabla\left(-\frac{1}{n}\sum_{i=1}^{n}\log p(y_i \mid x_i;\beta) + \frac{\gamma}{2}\|\beta\|_2^2\right)\\ &= -\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \sigma(x_i^T\beta)\right)x_i + \gamma\beta\end{aligned}$$

We rewrite it in vector form:

$$\nabla L(\beta) = -\frac{1}{n}X^T(y - \sigma(X\beta)) + \gamma\beta$$

[9]:
```
def compute_the_gradient(x, y, beta, gamma):
  return -1 / (y.shape[0]) * (x.T @ (y - torch.sigmoid(x @ beta))) + gamma *␣
↪beta
```

[10]:
```
# check output
torch.manual_seed(42)
gamma = 0.1

beta = torch.rand([X.shape[1], 1], dtype=torch.float32, requires_grad=True)
loss = compute_the_loss(X, Y, beta, gamma)
loss.backward()
auto_grad = beta.grad

m_gradient = compute_the_gradient(X, Y, beta, gamma)
torch.allclose(m_gradient, auto_grad)
```

[10]: True

[11]:
```
beta.shape
```

[11]: torch.Size([691, 1])

1.5.2 Problem 2.2: Implement Gradient Descent

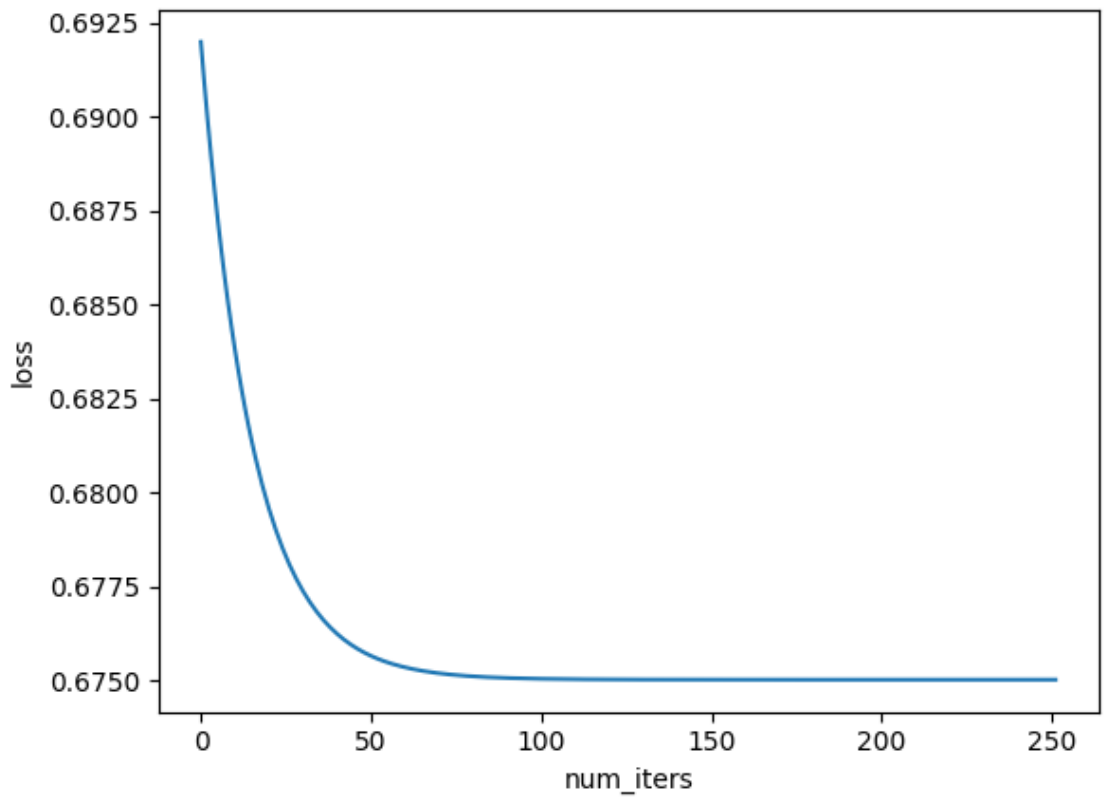
Now, use gradient descent to fit your Bradley-Terry model to the provided data.

Deliverables for this question: 1. Code the implements gradient descent to fit your Bradley-Terry model to the provided data. 2. A plot of the loss curve of your algorithm and a brief discussion if it makes sense or not 3. A plot of the histogram of the fitted values of β 4. The top 10 teams from your ranking, and a discussion of whether this ranking makes sense or not.

[12]:
```
# your code here (you can use multiple code and markdown cells to organize your␣
↪answer)
# 1. Code the implements gradient descent to fit your Bradley-Terry model to␣
↪the provided data.
gamma = 0.01
beta = torch.zeros([X.shape[1], 1], dtype=torch.float32, requires_grad=True)
lr = 3
num_iters = 1000

losses = []
for _ in range(num_iters):
  c_gradient = compute_the_gradient(X, Y, beta, gamma)

  with torch.no_grad():
    new_beta = beta - lr * c_gradient

    if torch.allclose(beta, new_beta, rtol=1e-4):
      break

    beta.copy_(new_beta)

  losses.append(compute_the_loss(X, Y, beta, gamma).item())
```

[13]:
```
# 2. A plot of the loss curve of your algorithm and a brief discussion if it␣
↪makes sense or not
plt.plot(torch.as_tensor(losses))
plt.xlabel("num_iters")
plt.ylabel("loss")
```

[13]:
```
Text(0, 0.5, 'loss')
```



[14]:
```
len(beta)
```

[14]: 691

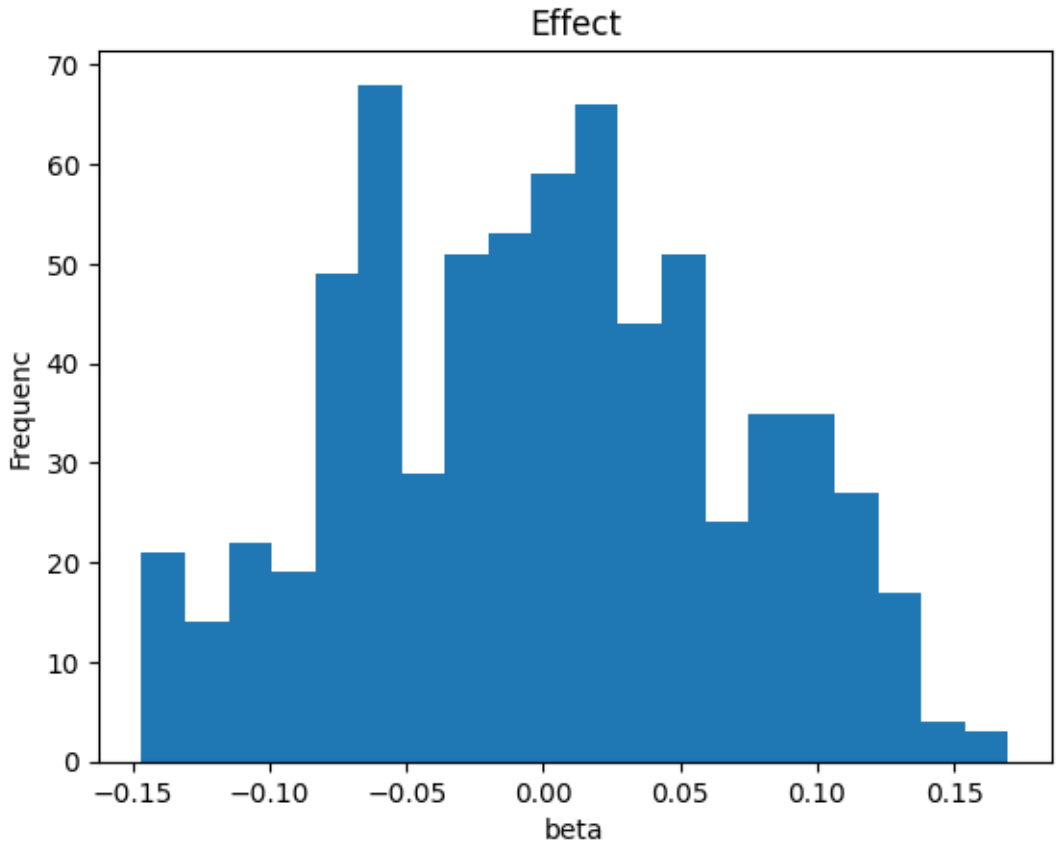
[15]:
```
df_beta = pd.DataFrame({"Team": teams, "Effect": beta.detach().numpy().
 ↪flatten()})
df_beta
```

[15]:

	Team	Effect
0	Abilene Christian	-0.014262
1	Adams State	-0.067192
2	Adrian	-0.106660
3	Air Force	0.053968
4	Akron	-0.107294
..	…	…
686	Worcester Polytechnic	-0.052772
687	Worcester State College	-0.094957
688	Wyoming	0.054157
689	Yale	0.053426
690	Youngstown State	0.040611

```
[691 rows x 2 columns]
```

[16]:
```
# 3. A plot of the histogram of the fitted values of $\beta$
plt.hist(df_beta["Effect"], bins=20) # $bins ==> k = \lceil \log_2(n) + 1␣
↪\rceil$
plt.title("Effect")
plt.xlabel("beta")
plt.ylabel("Frequenc")
plt.show()
```



[17]:
```
# 4. The top 10 teams from your ranking, and a discussion of whether this␣
↪ranking makes sense or not.
teams_10_gd = df_beta.sort_values(by="Effect", ascending=False).head(10)
teams_10_gd
```

[17]:

	Team	Effect
303	Liberty	0.169892
631	Washington	0.160030
347	Michigan	0.159504

```
558                Texas  0.147884
214              Georgia  0.147814
129       Colorado Mines  0.147378
196        Florida State  0.145838
509  South Dakota State  0.136141
271        Johns Hopkins  0.135715
5                Alabama  0.134950
```

1.6 Problem 3: Newton's Method

Now, use Newton's method to fit your Bradley-Terry model to the provided data.

1.6.1 Problem 3.1 The Hessian

Problem 3.1.1. Implement and check the Hessian Write a function to compute the Hessian of the average negative log likelihood and check your answer against the output of `from torch.autograd.functional.hessian`.

Our solution comes from Logistic Regression — STATS 305B_ Models and Algorithms for Discrete Data–Convexity of the Log Likelihood

Answer:

$$\nabla L(\beta) = -\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \sigma(x_i^T\beta)\right)x_i + \gamma\beta$$

We need compute the Hessian,

$$\nabla^2 L(\beta) = \frac{1}{n}\sum_{i=1}^{n}\sigma'(\beta^T x_i)x_i x_i^T + \gamma\mathbf{I}_{dim(\beta)}$$

We know $\sigma'(a) = \sigma(a)(1-\sigma(a))$, So we get

$$\begin{aligned}\nabla^2 L(\beta) &= \frac{1}{n}\sum_{i=1}^{n}\sigma(\beta^\top x_i)(1-\sigma(\beta^\top x_i))x_i x_i^\top + \gamma\mathbf{I}_{dim(\beta)} \\ &= \frac{1}{n}\sum_{i=1}^{n}w_i x_i x_i^\top + \gamma\mathbf{I}_{dim(\beta)}\end{aligned}$$

where $w = \sigma(\beta^\top x_i)(1-\sigma(\beta^\top x_i))$, We rewrite it in vector form:

$$\nabla^2 L(\beta) = \frac{1}{n}X^T W X + \gamma\mathbf{I}_{dim(\beta)}$$

where $W_{ij} = w_{ij}$, a diagonal matrix.

[18]:
```
beta.shape[0]
```

[18]: 691

[19]:
```
def compute_the_hessian(X, Y, beta, gamma):
  a = torch.sigmoid(X @ beta).squeeze()
```

```
    W = torch.diag(a * (1 - a))
    return (1 / Y.shape[0]) * (X.T @ W @ X) + gamma * torch.eye(beta.shape[0])
```

[20]:
```
# check output
torch.manual_seed(42)
gamma = 0.1

beta = torch.rand([X.shape[1], 1], dtype=torch.float32, requires_grad=True)
h_left = compute_the_hessian(X, Y, beta, gamma)

h_loss = lambda beta: compute_the_loss(X, Y, beta, gamma)
right_h = torch.autograd.functional.hessian(h_loss, beta).squeeze()

torch.allclose(h_left, right_h)
```

[20]: True

[21]:
```
h_left.shape[0], right_h.shape[0]
```

[21]: (691, 691)

Problem 3.1.2: Positive definiteness Compute the Hessian at the point $\beta = 0$ without regularization (set $\gamma = 0$). Unless you've done sort of pre-processing, it's probably singular.

[22]:
```
gamma=0
beta=torch.zeros([X.shape[1], 1], dtype=torch.float32, requires_grad=True)

h = compute_the_hessian(X, Y, beta, gamma)

print("rank: ", torch.linalg.matrix_rank(h))
print("Singular: ", torch.linalg.det(h))
```

```
rank:  tensor(687)
Singular:  tensor(-0., grad_fn=<LinalgDetBackward0>)
```

[23]:
```
h.shape[0] # not full rank
```

[23]: 691

Problem 3.1.3 Describe intuitively and mathematically what it means for the Hessian of the negative log likelihood to be singular in the context of this data and model

Answer: Mathematically, a Hessian singular matrix is non-invertible, has no unique solution, and undergoes a geometrical collapse. Our information is lost during this collapse process, making it impossible to deduce the original state.

Statistically, our rugby team could potentially have a perfect record or a perfect record.

Problem 3.1.4 Give a hypothesis for why the Hessian in this dataset and model is singular, and provide empirical evidence to support your hypothesis.

Answer: Our covariate matrix X is linearly correlated; it is not full rank.

[24]:
```
# your code here
print("rank: ", torch.linalg.matrix_rank(X)) # X not column full rank
```

```
rank:  tensor(687)
```

Problem 3.1.5 Explain why the Hessian is invertible when $\gamma > 0$.

Answer: When $\gamma > 0$, since our original Hessian matrix is positive semi-definite (≥ 0), γ will push all the eigenvalues upwards, and the matrix then becomes strictly positive definite (> 0).

1.6.2 Problem 3.2: Implement Newton's method

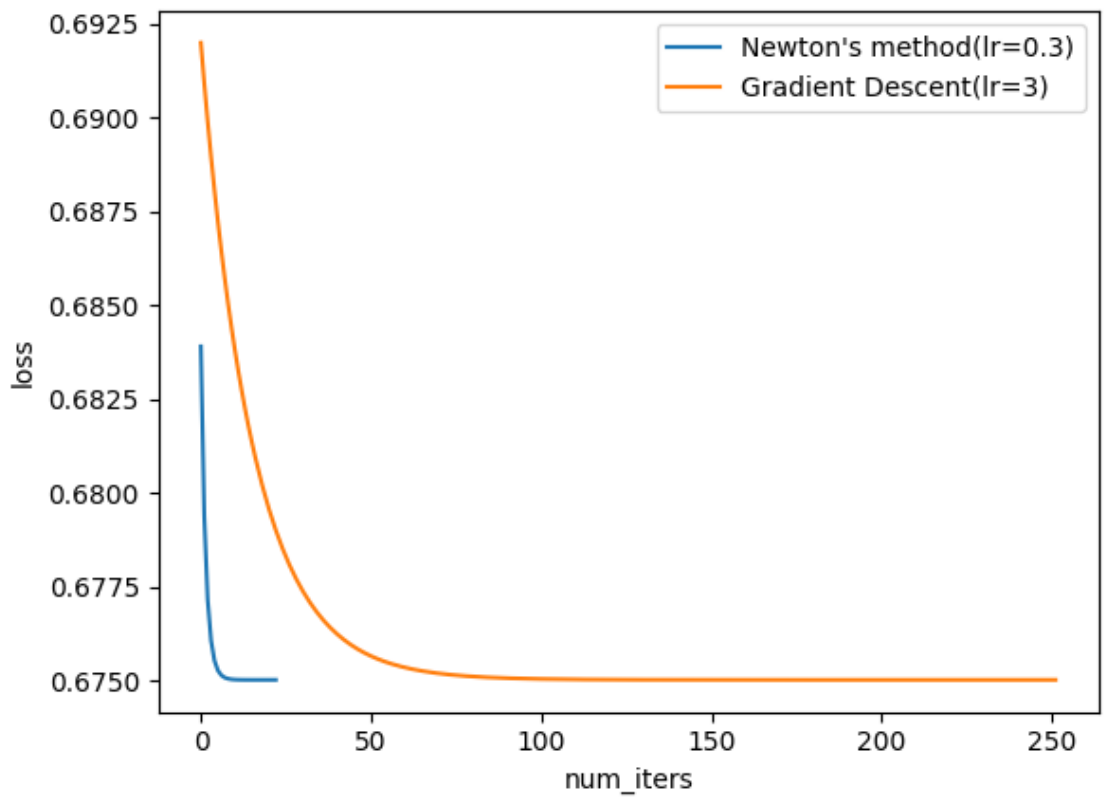
Now, use Newton's method to fit your ℓ_2-regularized Bradley-Terry model to the provided data.

Deliverables for this question: 1. Code the implements Newton's method to fit your Bradley-Terry model to the provided data. 2. A plot of the loss curves from Newton's method and from gradient descent, using the same regularization strength γ and initialization β_0. Briefly discuss the results and compare their rates of convergence. 3. A plot of the histogram of the fitted values of β 4. The top 10 teams from your ranking, and a discussion of whether this ranking makes sense or not.

[25]:
```
# your code here (you can use multiple code and markdown cells to organize your␣
↪answer)
# 1. Code the implements Newton's method to fit your Bradley-Terry model to the␣
↪provided data.
gamma = 0.01
beta = torch.zeros([X.shape[1], 1], dtype=torch.float32, requires_grad=True)
lr = 0.3
num_iters = 100

newton_losses = []
for _ in range(num_iters):
  hessian = compute_the_hessian(X, Y, beta, gamma)
  a_gradient = compute_the_gradient(X, Y, beta, gamma)

  with torch.no_grad():
    new_beta = beta - lr * torch.linalg.solve(hessian, a_gradient)

    if torch.allclose(beta, new_beta, rtol=1e-4):
      break

    beta.copy_(new_beta)

  beta, newton_losses.append(compute_the_loss(X, Y, beta, gamma).item())
```

[26]:
```
# 2. A plot of the loss curves from Newton's method and from gradient descent,
# using the same regularization strength $\gamma$ and initialization $\beta_0$.
# Briefly discuss the results and compare their rates of convergence.
plt.plot(torch.as_tensor(newton_losses), label="Newton's method(lr=0.3)")
plt.plot(torch.as_tensor(losses), label="Gradient Descent(lr=3)")
plt.xlabel("num_iters")
plt.ylabel("loss")
plt.legend()
```

[26]: <matplotlib.legend.Legend at 0x11462c440>



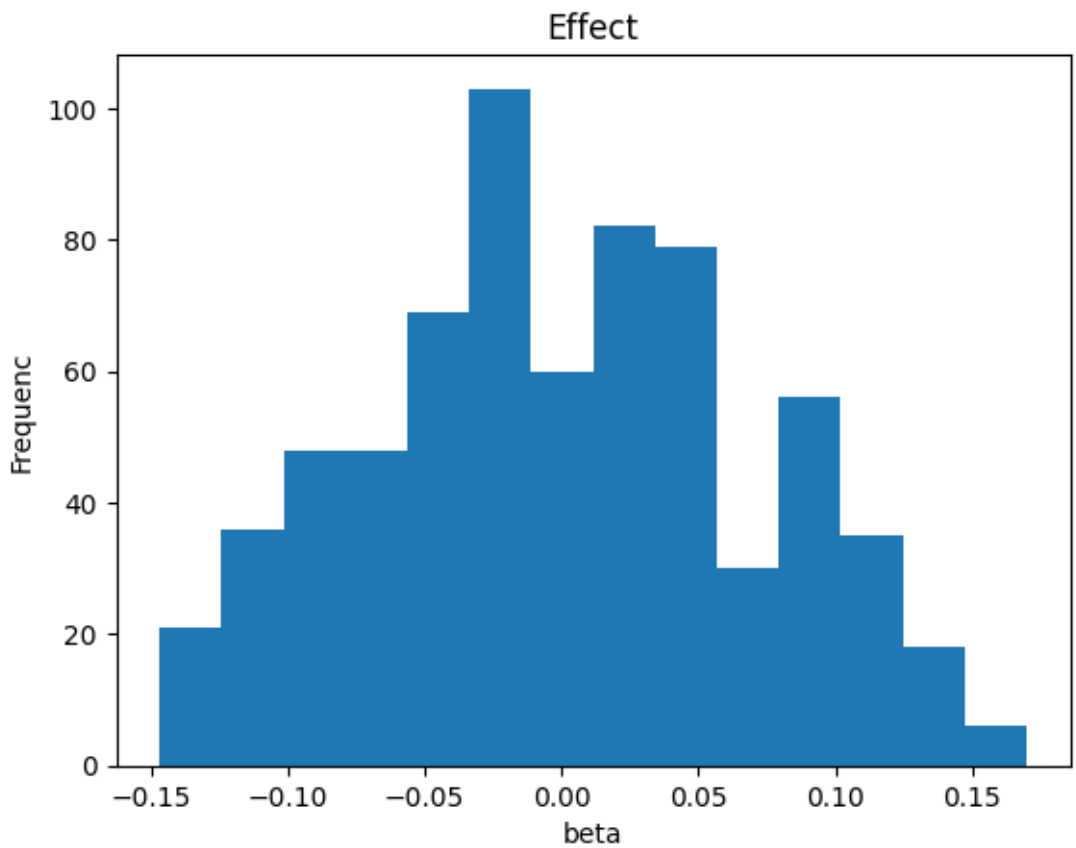
[27]:
```
df_newton_beta = pd.DataFrame({"Team": teams, "Effect": beta.detach().numpy().
 ↪flatten()})
df_newton_beta
```

[27]:

	Team	Effect
0	Abilene Christian	-0.014263
1	Adams State	-0.067192
2	Adrian	-0.106657
3	Air Force	0.053968
4	Akron	-0.107292

```
..                        …       …
686    Worcester Polytechnic -0.052770
687  Worcester State College -0.094959
688                  Wyoming  0.054158
689                     Yale  0.053424
690         Youngstown State  0.040611

[691 rows x 2 columns]
```

[28]:
```
# 3. A plot of the histogram of the fitted values of $\beta$.
plt.hist(df_newton_beta["Effect"], bins='auto') # $bins ==> k = \lceil␣
↪\log_2(n) + 1 \rceil$ or 'auto'
plt.title("Effect")
plt.xlabel("beta")
plt.ylabel("Frequenc")
plt.show()
```



[29]:
```
# 4. The top 10 teams from your ranking, and a discussion of whether this␣
↪ranking makes sense or not.
```

```
teams_10_n_gd = df_newton_beta.sort_values(by="Effect", ascending=False).
↪head(10)
teams_10_n_gd
```

[29]:

```
                   Team    Effect
303             Liberty  0.169881
631          Washington  0.160027
347            Michigan  0.159500
558               Texas  0.147884
214             Georgia  0.147814
129      Colorado Mines  0.147376
196       Florida State  0.145834
509  South Dakota State  0.136144
271       Johns Hopkins  0.135717
5               Alabama  0.134951
```

Therefore, let's compare the final values calculated by gradient descent and Newton's method.

[30]:

```
compare_ = pd.concat(
    [teams_10_gd, teams_10_n_gd],
    axis=1,
    keys=["GD", "Newton"]
)
compare_
```

[30]:

```
                     GD                        Newton
                   Team    Effect                 Team    Effect
303             Liberty  0.169892             Liberty  0.169881
631          Washington  0.160030          Washington  0.160027
347            Michigan  0.159504            Michigan  0.159500
558               Texas  0.147884               Texas  0.147884
214             Georgia  0.147814             Georgia  0.147814
129      Colorado Mines  0.147378      Colorado Mines  0.147376
196       Florida State  0.145838       Florida State  0.145834
509  South Dakota State  0.136141  South Dakota State  0.136144
271       Johns Hopkins  0.135715       Johns Hopkins  0.135717
5               Alabama  0.134950             Alabama  0.134951
```

1.7 Problem 4: Model criticism and revision

Let's take another look the Bradley-Terry model from earlier and think about improvements we can make.

1.7.1 Problem 4.1: Improvements to Bradley-Terry Model

Choose one way to improve the Bradley-Terry model. Discuss *a priori* why you think this change will improve the model and implement your change.

Answer:

We replace the original

$$Y_i \sim \text{Bern}(\sigma(\beta_{h(i)} - \beta_{a(i)}))$$

with

$$Y_i \sim \text{Bern}(\sigma(f_\theta(s_i) - f_\theta(s_j)))$$

where, $s_i = f_\theta(x_i)$, f_θ is the neural network, x_i is the team feature, and s_i is the scalar score.

Our parameter θ is shared, and our model requires input features. This type of model is called a Neural Bradley-Terry Model, and our implementation above uses pointwise scoring.

We should re-preprocess the data (each team $k \to$ one feature vector z_k)

However, after implementing the neural network version, our neural network caused the original standard model to lose its convex optimization properties. This meant that the final calculated value was a local optimum, making our results completely inconsistent with the standard model. Therefore, I decided to remove the neural network (MLP) and implement the model using only nn.Embedding + self.home_adv (home advantage, a custom parameter; since we removed the neutral site, we don't need to set it to 0). For our improved task, our self.home_adv is fully satisfactory, while maintaining interpretability.

[31]:
```
df_n = allgames.dropna(subset=["Home Points", "Away Points", "Neutral Site"])
df_n = df_n[df_n["Home Points"] != df_n["Away Points"]]
df_n = df_n[df_n["Neutral Site"] != True]

Y_N = torch.tensor((df_n["Home Points"] > df_n["Away Points"]).astype(int).
 ↪values, dtype=torch.float32).unsqueeze(1)

teams = sorted(set(df_n["Home Team"]) | set(df_n["Away Team"]))
team_idx = {team: i for i, team in enumerate(teams)}
T_N = len(teams)

df_n = df_n.reset_index(drop=True)

home_idx = []
away_idx = []

for _, r in df_n.iterrows():
  home_idx.append(team_idx[r["Home Team"]])
  away_idx.append(team_idx[r["Away Team"]])

home_idx = torch.tensor(home_idx, dtype=torch.long)
away_idx = torch.tensor(away_idx, dtype=torch.long)
```

[32]:
```
home_idx.shape, away_idx.shape
```

[32]:
```
(torch.Size([3469]), torch.Size([3469]))
```

[33]:
```
# Model Implement
class BT(nn.Module):
```

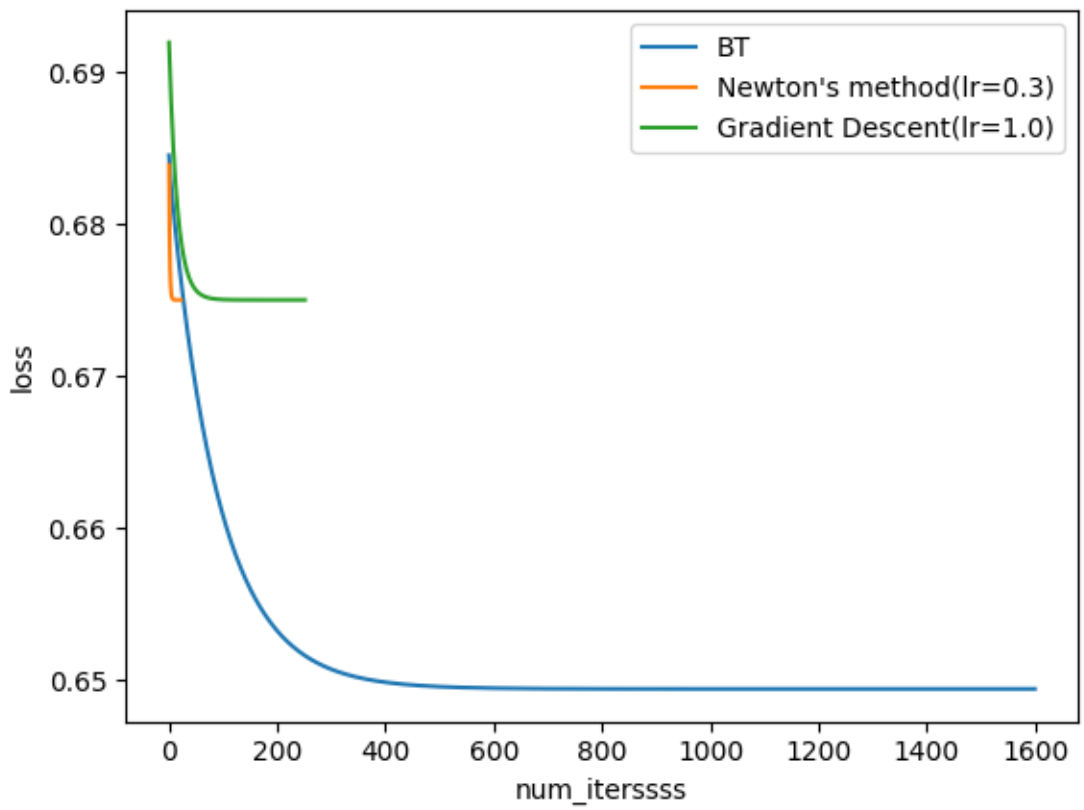
```
  def __init__(self, num_teams):
        super().__init__()
        self.strength = nn.Embedding(num_teams, 1)

        nn.init.zeros_(self.strength.weight)

        self.home_adv = nn.Parameter(torch.tensor([0.2]))

  def forward(self, h_idx, a_idx):
        s_i = self.strength(h_idx)
        s_j = self.strength(a_idx)

        return s_i - s_j + self.home_adv
```

[34]:
```
model = BT(num_teams=T_N)
loss_f = nn.BCEWithLogitsLoss()

optimizer = optim.SGD(model.parameters(), lr=1.0, weight_decay=1e-2)
losses_epoch = []
num_iters = 1600

model.train()
for _ in range(1600):
    optimizer.zero_grad()

    logits = model(home_idx, away_idx)

    loss = loss_f(logits, Y_N)

    loss.backward()

    optimizer.step()

    losses_epoch.append(loss.item())

print(f"Final train-loss: {losses_epoch[-1]:.4f}")
```

```
Final train-loss: 0.6494
```

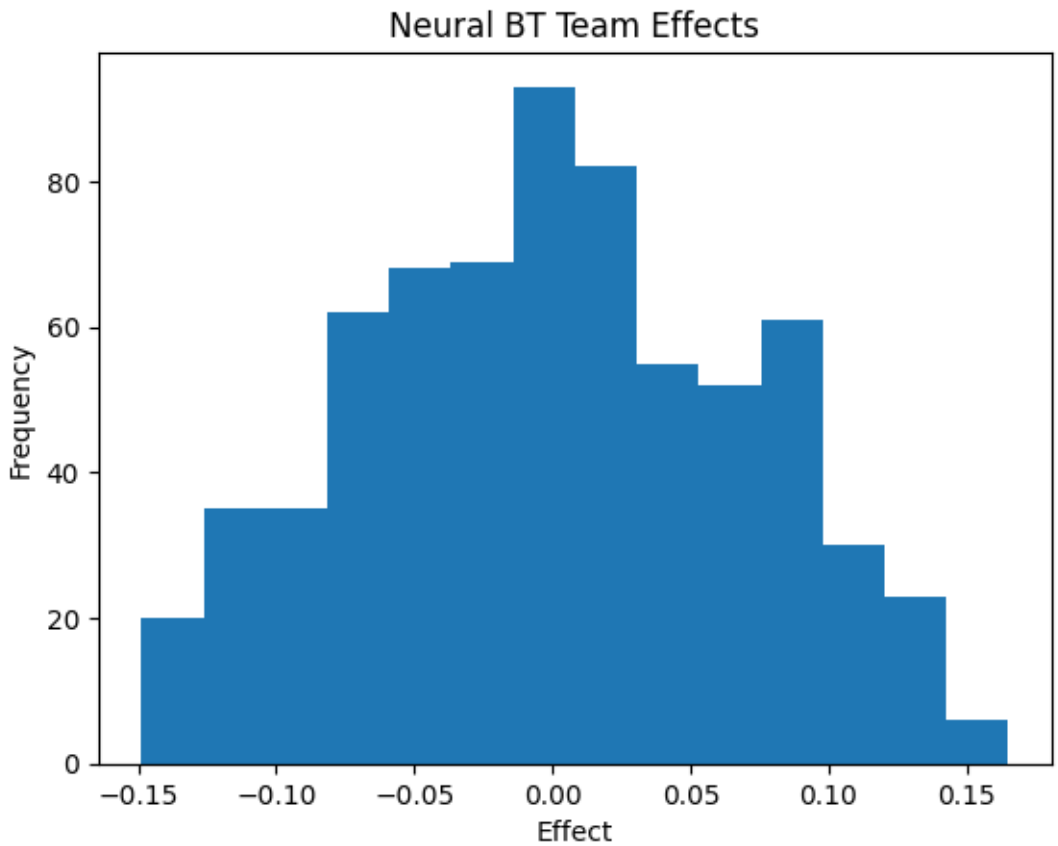
[35]:
```
plt.plot(losses_epoch, label="BT")
plt.plot(torch.as_tensor(newton_losses), label="Newton's method(lr=0.3)")
plt.plot(torch.as_tensor(losses), label="Gradient Descent(lr=1.0)")
plt.xlabel("num_iterssss")
plt.ylabel("loss")
plt.legend()
```

[35]:
```
<matplotlib.legend.Legend at 0x11a86a990>
```



[36]:
```
with torch.no_grad():
    S = model.strength.weight.squeeze().numpy()
    S = S - S.mean()
```

[37]:
```
df_nbt_beta = pd.DataFrame({"Team": teams, "Effect": S})

plt.hist(df_nbt_beta["Effect"], bins='auto')
plt.title("Neural BT Team Effects")
plt.xlabel("Effect")
plt.ylabel("Frequency")
plt.show()
```



[38]:
```
# top-10
teams_10_nbt = df_nbt_beta.sort_values(by="Effect", ascending=False).head(10)
teams_10_nbt
```

[38]:

	Team	Effect
303	Liberty	0.165172
631	Washington	0.156707
347	Michigan	0.156175
558	Texas	0.146129
129	Colorado Mines	0.145859
196	Florida State	0.144211
214	Georgia	0.142652
271	Johns Hopkins	0.135894
549	Susquehanna	0.134814
436	Ohio State	0.134389

[39]:
```
compare_ = pd.concat(
    [
      teams_10_gd.reset_index(drop=True),
      teams_10_n_gd.reset_index(drop=True),
```

```
      teams_10_nbt.reset_index(drop=True),
    ],
    axis=1,
    keys=["GD", "Newton", "SGD+Home Advantage"]
)
compare_.head(10)
```

[39]:

```
                    GD                        Newton            \
                  Team    Effect                 Team    Effect
0             Liberty  0.169892             Liberty  0.169881
1          Washington  0.160030          Washington  0.160027
2            Michigan  0.159504            Michigan  0.159500
3               Texas  0.147884               Texas  0.147884
4             Georgia  0.147814             Georgia  0.147814
5      Colorado Mines  0.147378      Colorado Mines  0.147376
6       Florida State  0.145838       Florida State  0.145834
7  South Dakota State  0.136141  South Dakota State  0.136144
8       Johns Hopkins  0.135715       Johns Hopkins  0.135717
9             Alabama  0.134950             Alabama  0.134951

  SGD+Home Advantage
                 Team    Effect
0            Liberty  0.165172
1         Washington  0.156707
2           Michigan  0.156175
3              Texas  0.146129
4     Colorado Mines  0.145859
5      Florida State  0.144211
6            Georgia  0.142652
7      Johns Hopkins  0.135894
8        Susquehanna  0.134814
9         Ohio State  0.134389
```

1.7.2 Problem 4.2: Evaluation

Assess whether or not your change was an improvement or not. Provide empirical evidence by evaluating performance on a held out test set and include at least one plot supporting your assessment.

Answer: We will use different splitting methods to split the training/test set (all other parameters are the same).

[40]:

```
from sklearn.model_selection import train_test_split

##########################################################################################
idx = np.arange(len(home_idx))

train_idx, test_idx = train_test_split(idx, test_size=0.2, random_state=42)
```

```
train_home = home_idx[train_idx]
train_away = away_idx[train_idx]
tarin_y = Y_N[train_idx]

test_home = home_idx[test_idx]
test_away = away_idx[test_idx]
test_y = Y_N[test_idx]
##########################################################################################

x_train, x_test, y_train, y_test = train_test_split(X, Y, test_size=0.2,␣
↪random_state=42)
```

[41]:
```
def evaluate_pytorch_model(model, home_idx, away_idx, target_y):
  model.eval()

  with torch.no_grad():

    logits = model(home_idx, away_idx)

    # If logits > 0, it means the model predicts the home team will win.
    preds = (logits >= 0).float()

    accuracy = (preds == target_y).float().mean()

    return accuracy.item()

def evaluate_origin_bradley_terry(X, Y, beta):
  with torch.no_grad():

    logits = X @ beta

    preds = (logits >= 0).float()

    accuracy = (preds == Y).float().mean()

    return accuracy.item()
```

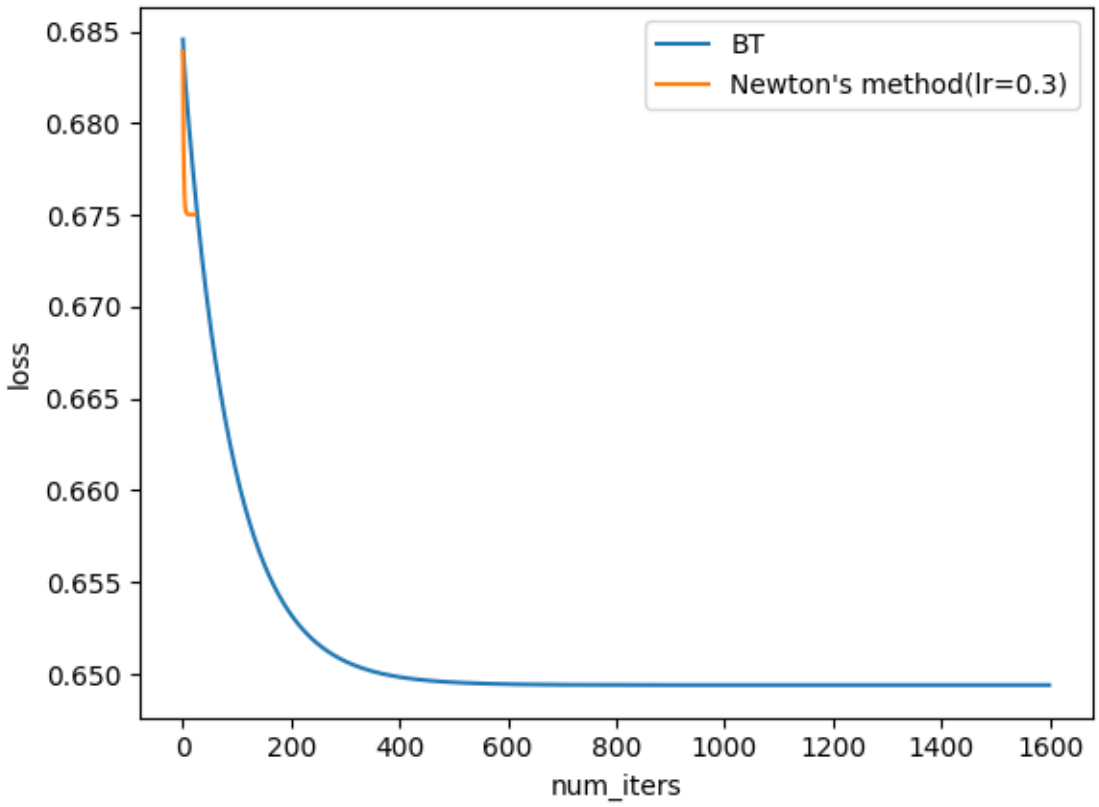
[42]:
```
train_bn_acc = evaluate_pytorch_model(model, train_home, train_away, tarin_y)
test_bn_acc = evaluate_pytorch_model(model, test_home, test_away, test_y)

train_acc = evaluate_origin_bradley_terry(x_train, y_train, beta)
test_acc = evaluate_origin_bradley_terry(x_test, y_test, beta)

print(f"Improved Model -> Train Acc: {train_bn_acc:.4f}, Test Acc: {test_bn_acc:
↪.4f}")
print(f"Origin Model -> Train Acc: {train_acc:.4f}, Test Acc: {test_acc:.4f}")
```

```
Improved Model -> Train Acc: 0.5647, Test Acc: 0.5836
```

```
Origin Model -> Train Acc: 0.8468, Test Acc: 0.8386
```

[43]:
```
plt.plot(losses_epoch, label="BT")
plt.plot(torch.as_tensor(newton_losses), label="Newton's method(lr=0.3)")
plt.xlabel("num_iters")
plt.ylabel("loss")
plt.legend()
```

[43]:
```
<matplotlib.legend.Legend at 0x11f7442d0>
```



It turns out, by improving our model, we reduced our training loss (compared to the original model). However, a counterintuitive situation arose: the accuracy was far lower than that of our original model, and the improved version even achieved higher test accuracy than training accuracy. This forced us to consider the reasons: was it due to ambiguity in the data or a problem with the model?

1.7.3 Problem 4.3: Reflection

Reflecting on the analysis we've conducted in this assignemnt, which conference is best? Is there a significant difference? Please justify your answer.

Answer: We will

[44]:
```
df.head().T
```

[44]:
```
                                        0                          1  \
Id                                401520145                  401525450
Season                                 2023                       2023
Week                                      1                          1
Season Type                         regular                    regular
Start Date          2023-08-26T21:30:00.000Z  2023-08-26T23:00:00.000Z
Start Time Tbd                        False                      False
Completed                              True                       True
Neutral Site                          False                      False
Conference Game                        True                      False
Attendance                          17982.0                    15356.0
Venue Id                             3662.0                     3601.0
Venue                            JSU Stadium    Aggie Memorial Stadium
Home Id                                 55                        166
Home Team                 Jacksonville State          New Mexico State
Home Conference               Conference USA            Conference USA
Home Division                            fbs                        fbs
Home Points                             17.0                       30.0
Home Line Scores                         NaN                        NaN
Home Post Win Prob                  0.408001                   0.239249
Home Pregame Elo                     1500.0                     1322.0
Home Postgame Elo                    1499.0                     1274.0
Away Id                                2638                        113
Away Team                              UTEP                      UMass
Away Conference               Conference USA          FBS Independents
Away Division                            fbs                        fbs
Away Points                             14.0                       41.0
Away Line Scores                         NaN                        NaN
Away Post Win Prob                  0.591999                   0.760751
Away Pregame Elo                     1369.0                     1074.0
Away Postgame Elo                    1370.0                     1122.0
Excitement Index                    6.821333                   5.311493
Highlights                               NaN                        NaN
Notes                                    NaN                        NaN

                                        2                          3  \
Id                                401532392                  401540628
Season                                 2023                       2023
Week                                      1                          1
Season Type                         regular                    regular
Start Date          2023-08-26T23:00:00.000Z  2023-08-26T23:00:00.000Z
Start Time Tbd                        False                      False
Completed                              True                       True
Neutral Site                          False                      False
Conference Game                       False                      False
```

Attendance	23867.0	NaN
Venue Id	7311.0	4419.0
Venue	Snapdragon Stadium	Bob Ford Field
Home Id	21	399
Home Team	San Diego State	Albany
Home Conference	Mountain West	CAA
Home Division	fbs	fcs
Home Points	20.0	34.0
Home Line Scores	NaN	NaN
Home Post Win Prob	0.954469	0.922517
Home Pregame Elo	1488.0	NaN
Home Postgame Elo	1497.0	NaN
Away Id	195	2230
Away Team	Ohio	Fordham
Away Conference	Mid-American	Patriot
Away Division	fbs	fcs
Away Points	13.0	13.0
Away Line Scores	NaN	NaN
Away Post Win Prob	0.045531	0.077483
Away Pregame Elo	1482.0	NaN
Away Postgame Elo	1473.0	NaN
Excitement Index	6.547378	5.608758
Highlights	NaN	NaN
Notes	NaN	NaN

	4
Id	401520147
Season	2023
Week	1
Season Type	regular
Start Date	2023-08-26T23:30:00.000Z
Start Time Tbd	False
Completed	True
Neutral Site	False
Conference Game	False
Attendance	21407.0
Venue Id	3973.0
Venue	Vanderbilt Stadium
Home Id	238
Home Team	Vanderbilt
Home Conference	SEC
Home Division	fbs
Home Points	35.0
Home Line Scores	NaN
Home Post Win Prob	0.180846
Home Pregame Elo	1322.0
Home Postgame Elo	1327.0

```
Away Id                                   62
Away Team                           Hawai'i
Away Conference                Mountain West
Away Division                            fbs
Away Points                             28.0
Away Line Scores                         NaN
Away Post Win Prob                  0.819154
Away Pregame Elo                      1246.0
Away Postgame Elo                     1241.0
Excitement Index                    5.282033
Highlights                               NaN
Notes                                    NaN
```

[45]:
```
home_con = df[["Home Team", "Home Conference"]].rename(columns={"Home Team":
 ↪"Team", "Home Conference":"Conference"})
away_con = df[["Away Team", "Away Conference"]].rename(columns={"Away Team":
 ↪"Team", "Away Conference":"Conference"})
team_to_con = pd.concat([home_con, away_con], ignore_index=True).
 ↪drop_duplicates().set_index("Team")["Conference"].to_dict()
```

[46]:
```
from itertools import islice

for i, (k, v) in enumerate(islice(team_to_con.items(), 5)):
    print(i, k, v)
```

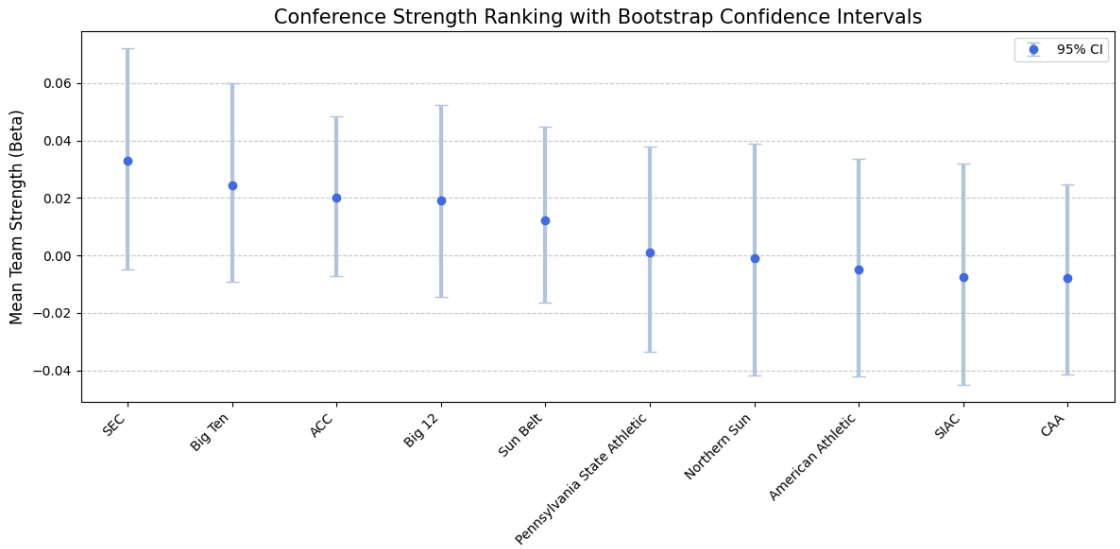
```
0 Jacksonville State Conference USA
1 New Mexico State Conference USA
2 San Diego State Mountain West
3 Albany CAA
4 Vanderbilt SEC
```

[47]:
```
# Bootstrap Sampling
df_newton_beta["Conference"] = df_newton_beta["Team"].map(team_to_con)
num_iters = 1000
bootstrap_data = []

# nlargest() -> Retrieves the top 12 most representative conferences
# value_counes() -> Counts the number of teams in each conference
# index -> Gets a list of these conference names
top_conferences = df_newton_beta["Conference"].value_counts().nlargest(10).index
```

[48]:
```
for conf in top_conferences:
    conf_effects = df_newton_beta[df_newton_beta["Conference"] ==␣
 ↪conf]["Effect"].values

    if len(conf_effects) > 0:
        resampled_means = []
```

```
        for _ in range(num_iters):
            # Sampling with replacement from the current conference's effect␣
↪values generates a new sample of the same length
            sample = np.random.choice(conf_effects, size=len(conf_effects),␣
↪replace=True)
            # Looping 1000 times => yields 1000 averages =>
            # approximating the sampling distribution of the average effect of␣
↪this conference
            resampled_means.append(np.mean(sample))

        # Record the mean and 95% confidence interval
        # Low / High: 2.5% and 97.5% quantiles => 95% bootstrap confidence␣
↪interval
        bootstrap_data.append({
            "Conference": conf,
            "Mean": np.mean(resampled_means),
            "Low": np.percentile(resampled_means, 2.5),
            "High": np.percentile(resampled_means, 97.5)
        })
```

[49]:
```
# Visualization: Draw an error bar chart with confidence intervals
df_boot = pd.DataFrame(bootstrap_data).sort_values(by='Mean', ascending=False)
plt.figure(figsize=(12, 6))
plt.errorbar(
    df_boot['Conference'],
    df_boot['Mean'],
    yerr=[df_boot['Mean'] - df_boot['Low'], df_boot['High'] - df_boot['Mean']],␣
↪# Compute 95% confidence interval
    fmt='o', color='royalblue', ecolor='lightsteelblue', elinewidth=3,␣
↪capsize=5, label='95% CI'
)

# Tilt the label at a 45-degree angle to avoid crowding, Right-aligned text on␣
↪the horizontal axis
plt.xticks(rotation=45, ha='right')

plt.title("Conference Strength Ranking with Bootstrap Confidence Intervals",␣
↪fontsize=15)
plt.ylabel("Mean Team Strength (Beta)", fontsize=12)
plt.grid(axis='y', linestyle='--', alpha=0.7) # Add horizontal grid
plt.legend()
plt.tight_layout() # Automatic layout adjustment
plt.show()
```



[50]:
```python
# Print the numerical results for Reflection to use
print("Bootstrap statistics (top 10):")
print(df_boot.head(10)[['Conference', 'Mean', 'Low', 'High']])
```

```
Bootstrap statistics (top 10):
                    Conference      Mean       Low      High
8                          SEC  0.032804 -0.004969  0.072054
6                      Big Ten  0.024548 -0.009026  0.059780
4                          ACC  0.020241 -0.007121  0.048391
2                       Big 12  0.019073 -0.014456  0.052340
5                     Sun Belt  0.012127 -0.016465  0.044824
0  Pennsylvania State Athletic  0.001176 -0.033369  0.037889
3                  Northern Sun -0.000934 -0.041574  0.038712
7             American Athletic -0.004774 -0.042059  0.033745
9                         SIAC -0.007587 -0.045146  0.031986
1                          CAA -0.007883 -0.041292  0.024712
```

By observing the 95% Confidence Intervals and Low values in the above graph, we found that all Low values are negative (the Low value of Top 1 crosses 0) and the Confidence Intervals highly overlap (meaning we cannot conclude that Top 1 is stronger than its competitors, nor can we confirm with 95 % confidence that it is better than the entire league). Although SEC ranks first, we found that this result is not statistically significant.

1.8 Submission Instructions

Formatting: check that your code does not exceed 80 characters in line width. You can set *Tools → Settings → Editor → Vertical ruler column* to 80 to see when you've exceeded the limit.

Converting to PDF The simplest way to convert to PDF is to use the "Print to PDF" option in your browser. Just make sure that your code and plots aren't cut off, as it may not wrap lines.

Alternatively You can download your notebook in .ipynb format and use the following commands to convert it to PDF. Then run the following command to convert to a PDF:

```
jupyter nbconvert --to pdf <yourlastname>_hw<number>.ipynb
```

(Note that for the above code to work, you need to rename your file `<yourlastname>_hw<number>.ipynb`)

Installing nbconvert:

If you're using Anaconda for package management,

```
conda install -c anaconda nbconvert
```

Upload your .pdf file to Gradescope. Please tag your questions correctly! I.e., for each question, all of and only the relevant sections are tagged.

Please post on Ed or come to OH if there are any other problems submitting the HW.